



I-AIR: intention-aware travel itinerary recommendation via multi-signal fusion and spatiotemporal constraints

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Abstract

Personalized travel itinerary recommendation aims to suggest ordered sequences of points of interest (POIs) that align with user preferences while respecting temporal and spatial constraints. Although traditional methods have made progress in next-POI prediction and route planning, they often rely on static user preferences and overly simplistic spatial assumptions, overlooking contextual variability and non-linear distance effects. Additionally, many approaches decouple POI selection from itinerary construction and depend heavily on limited behavioral signals, failing to capture richer feedback such as ratings, dwell times, or click histories. To address these limitations, we propose an intention-aware deep learning framework that integrates diverse user signals into a unified itinerary planning model. A novel fusion-aware encoder assimilates both explicit and implicit user feedback to uncover latent preferences driving POI choices. The model combines a multi-head self-attention transformer to capture the sequential and temporal dynamics of user behavior, with a graph convolutional network (GCN) that models complex co-visitation patterns among POIs. These components are jointly trained to select optimal POIs and construct itineraries that respect user-specific constraints and time budgets. Our custom itinerary optimization heuristic balances user interest with travel and waiting costs, directly aligning recommendation with practical route planning. Extensive experiments on multiple real-world datasets - including theme parks and urban tourism - demonstrate that our model consistently outperforms state-of-the-art baselines across a range of metrics. It produces more coherent and temporally balanced itineraries, effectively reducing queue times and improving user satisfaction. Our results highlight the value of multi-signal fusion and contextual modeling in capturing nuanced user intent, setting a new benchmark for personalized itinerary generation.

Keywords Natural language processing · Personalized itinerary recommendation · Intention-aware recommendation · Graph neural networks · Artificial intelligence applications

1 Introduction

Recommending personalized itineraries - ordered sequences of attractions or Points of Interest (POIs) for users to visit - is of great importance in tourism and smart city applications (Chen et al. 2014; Gavalas et al. 2015; Zheng and

Xie 2011). Successful itinerary recommendation systems can enhance travel experiences by saving time and tailoring routes to individual preferences. This problem is closely related to next-POI recommendation (Yu et al. 2015; Jiang et al. 2016), travel planning under time and cost constraints (Zhang et al. 2015, 2016), and personalized route recommendation in urban environments (Quercia et al. 2014; Galbrun et al. 2016). Despite growing interest, providing high-quality itinerary recommendations remains challenging due to several key factors.

First, user interests are dynamic and context-dependent. For example, a user's enthusiasm for museums versus restaurants may fluctuate with the time of day or day of week; interest in restaurants typically peaks at meal times (Huang et al. 2019). Many existing models assume time-invariant or recently-static user preferences, which fails to capture such temporal context shifts (Halder et al. 2022). Second, spatial

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influence on POI choice is non-linear. Early approaches often modeled distance effects in a simple way (e.g. assuming visitation likelihood decays linearly or with a fixed radius) (Ye et al. 2011), which can be inadequate. In reality, users might travel farther for especially interesting places, and distance decay can follow complex patterns (e.g. power-law or cubic) (Chang et al. 2020). Ignoring non-linear distance dependencies can misrepresent user willingness to visit distant but worthwhile POIs. Third, POI recommendation and itinerary construction have largely been treated as separate problems (Lim et al. 2017; Halder et al. 2022). Most pipelines first generate a set of top- k POIs for the user and then apply a route optimization algorithm to arrange them. This two-stage decoupling can lead to suboptimal itineraries - for instance, recommending several attractive POIs that are far apart or all open at conflicting times, such that no feasible tour exists within the user's time budget (Lim et al. 2017; Halder et al. 2022). Ideally, candidate selection should be aware of scheduling constraints and vice versa. Fourth, and central to this work, existing recommender systems rarely leverage the full spectrum of user feedback signals. Most POI recommenders rely on implicit check-in data or sequences of visits as the sole evidence of user preference (Liu et al. 2016; Feng et al. 2015). Other signals such as explicit ratings/reviews, click histories on travel sites, dwell times at locations, and aggregate trajectory patterns are typically ignored. These underutilized signals could provide valuable clues to a user's intentions - for example, a high rating or long dwell time at a particular museum reveals a strong interest in that category, and frequently clicking on nightlife venues suggests an implicit attraction even if not yet visited. No prior itinerary recommendation work, to our knowledge, has jointly modeled explicit and implicit feedback of this variety.

To address these challenges, we propose an intention-aware itinerary recommendation model that integrates multi-source user signals and jointly optimizes POI selection and route construction. Specifically, our architecture consists of (1) a Transformer-based sequential encoder to capture temporal patterns in user behavior, (2) a graph convolutional network (GCN) to model POI relationships, and (3) a greedy itinerary planner that balances user interest and travel constraints. Our approach has several novel aspects and contributions:

- We propose a unique method for jointly modeling explicit feedback (e.g., ratings and preferences) and implicit signals (e.g., visit sequences, dwell times) to better capture user intentions. This fusion mechanism allows for a richer and more dynamic representation of user preferences, making our model more effective at adapting to evolving user interests.
- We introduce an intention-aware deep learning framework that integrates a multi-head attention-based Trans-

former with a graph convolutional network (GCN). This architecture not only models temporal preferences but also captures the spatial dependencies between POIs, significantly improving the quality of POI recommendations and the overall itinerary generation process.

- Unlike existing models that treat POI recommendation and itinerary construction as two separate tasks, our model combines these into a unified, end-to-end system. We use a greedy itinerary construction algorithm that selects and arranges POIs by optimizing both user satisfaction and time feasibility, ensuring that the final itinerary is both personalized and practical.
- We conduct extensive experiments on real-world datasets, including theme park and urban tourism scenarios, demonstrating that our model consistently outperforms existing state-of-the-art methods across multiple evaluation metrics. These results highlight the model's potential to support more intelligent, user-centric itinerary planning in practical applications.

The remainder of the paper is organized as follows. In Section 2, we review recent advances in itinerary recommendation, intention modeling, and multi-signal fusion in recommender systems. Section 3 defines the personalized itinerary recommendation task and introduces our proposed intention-aware framework, detailing each component with formal mathematical formulations and algorithms for user intent estimation, POI ranking, and itinerary construction. Section 4 presents our experimental setup, including datasets, baseline models, evaluation metrics, and comprehensive quantitative and qualitative results. Finally, Section 5 concludes the paper and discusses future directions for integrating user modeling with adaptive itinerary generation in real-world tourism applications.

2 Related work

2.1 POI recommendation and next-POI prediction

Traditional approaches Early point-of-interest recommendation techniques adapted collaborative filtering and matrix factorization to location-based data. For example, Ye et al. integrated geographical distance decay into user-based collaborative filtering, showing that spatial proximity can improve POI prediction (Ye et al. 2011). Similarly, Lian et al. proposed a joint geographical matrix factorization model (GeoMF) that factors in distance and user-POI preference to improve accuracy (Lian et al. 2014). Such methods established that accounting for spatial context is crucial for POI recommendation. Another line of work incorporated Markov chains to capture sequential check-in patterns. Feng

et al. developed a personalized ranking metric embedding (PRME) model that extends factorization with a transition component, effectively modeling successive POI visit behavior in a metric space (Feng et al. 2015). These pre-deep-learning approaches, while innovative, often relied on linear assumptions or simple distance kernels to model geographical influence and did not fully exploit complex temporal or content information.

Deep sequential models The advent of deep learning brought more powerful sequence modeling to next-POI recommendation. Recurrent Neural Networks (RNNs) and their variants have been widely applied. Liu et al. introduced ST-RNN, which extends a time-dependent transition matrix within an RNN to capture spatio-temporal contexts of check-ins (Liu et al. 2016). Subsequent works added attention mechanisms and other enhancements. Huang et al. propose ATST-LSTM, an attention-based LSTM that attends to relevant historical check-ins for the current time slot (Huang et al. 2019). This model addresses the fact that a user's next move depends not only on the most recent visit but also on time-sensitive preferences (e.g. weekday vs weekend patterns). Similarly, attention networks and transformer architectures have been applied to POI sequence prediction. Chang et al. design a GCN-based model (GGLR) to learn geographical latent representations of POIs and integrate them with RNN outputs (Chang et al. 2020). Other approaches like the content-aware hierarchical embedding model by Chang et al. incorporate textual and category information of POIs into the sequential recommendation process (Li 2021). These methods demonstrate improved next-POI accuracy by combining sequence signals with auxiliary data such as POI descriptions or tags.

Graph and social context Beyond sequential patterns, researchers have looked at graph-based and social context for POI recommendation. Graph convolutional networks have shown success in modeling complex relationships in user-POI interaction graphs. For instance, the GGLR model by Chang et al. uses a graph neural network to capture higher-order connections between POIs (like co-visitation and region influence) (Chang et al. 2020). Yang et al. present a neural approach bridging collaborative filtering with semi-supervised learning on a POI graph, which leverages social check-in networks to propagate preference information (Yang et al. 2017). This helps address data sparsity by learning from both user feedback and unlabeled data (e.g. potential POIs with similar visitors). Likewise, Cheng et al. in their unified POI recommendation framework combine matrix factorization with geographically-driven regularization to meld collaborative signals and spatial clustering (Cheng et al. 2016). On the social side, many works exploit friends' check-ins or city-

level popularity. However, social influence is often modest compared to spatiotemporal influence (Ye et al. 2011), so recent studies emphasize more direct signals of user intent.

Multi-modal and advanced techniques Several recent studies integrate additional modalities or advanced training strategies for POI recommendation. Chen et al. incorporate textual reviews and POI descriptions via a deep collaborative filtering model, improving recommendations by understanding semantic aspects of POIs (Chen et al. 2020). Wang et al. propose STaTRL, a model that learns joint spatial-temporal and textual representations, showing that fusing POI content (e.g. tags, reviews) with visit sequences enhances performance (Wang et al. 2023). Ou et al. develop STA-TCN, which uses a spatial-temporal attention mechanism over a temporal CNN backbone for next-POI prediction (Ou et al. 2023). This captures short-term transition patterns with CNNs while using attention to emphasize important locations in the sequence. There is also growing interest in self-supervised learning and contrastive objectives: Kuo et al. introduce BERT-Trip, a contrastive learning framework that uses a BERT-based encoder to learn effective representations of entire trip sequences (Kuo et al. 2023). By creating augmented versions of trips and maximizing agreement, it learns embeddings that can be used for recommendation or itinerary retrieval.

Furthermore, issues of fairness, privacy, and scalability have started to be addressed in POI recommenders. Halder et al. incorporate capacity fairness by balancing popular and less-popular POIs in recommendations (Halder et al. 2023) - an aspect especially relevant when many users vie for limited-capacity attractions (e.g. theme park rides). Perifanis et al. present FedPOIRec, a federated learning approach to POI recommendation that preserves user privacy by training models across decentralized data silos (Perifanis et al. 2023). Long et al. (2023) propose a decentralized collaborative learning framework to reduce reliance on any single server and mitigate data silos for next-POI prediction (Long et al. 2023). These developments, while tangential to the core accuracy goal, indicate the maturing of POI recommendation research into practical deployment considerations.

In summary, next-POI recommendation has evolved from basic MF and Markov models to sophisticated deep models incorporating sequences, content, graphs, and self-supervision. However, most of these works concentrate on recommending a single next location or a set of POIs, rather than producing an entire ordered itinerary. They also typically utilize one primary source of feedback (check-in or rating data). Our work differs by focusing on the itinerary recommendation problem and by fusing multiple feedback signals to capture user intention more holistically.

2.2 Itinerary recommendation and tour planning

Early trip planning algorithms The itinerary recommendation problem can be seen as an instance of the Tourist Trip Design Problem (TTDP) or the Orienteering Problem with Time Windows, which are classic NP-hard optimization problems in operations research (Vansteenwegen et al. 2011). Early solutions applied heuristic or brute-force algorithms to generate sightseeing tours. Gavalas et al. developed the eCOMPASS system, which uses heuristic search to plan multi-modal tourist tours given user-specified preferences and time constraints (Gavalas et al. 2015). Similarly, a rich body of work exists on optimal route planning: for example, Vansteenwegen et al. survey various heuristic methods for orienteering (selecting a subset of locations to maximize score within a time limit) (Vansteenwegen et al. 2011). These methods typically require an interest score for each POI (which might be derived from popularity or user profile) and then focus on route optimization. While effective on small scales, pure optimization approaches struggle with personalized preference estimation and often assume the scores are given.

Some early data-driven itinerary recommenders mined social media and crowd-contributed data to infer popular routes. De Choudhury et al. is a seminal work that constructed travel itineraries by mining tagged geo-temporal “breadcrumbs” (e.g. Flickr photo timestamps and locations) to find commonly traveled sequences (De Choudhury et al. 2010). Brilhante et al. introduced TripBuilder, which plans touristic itineraries by leveraging user-generated photos and tags to determine points of interest and using a combination of clustering and orienteering algorithms to build tours (Brilhante et al. 2013). These approaches were among the first to demonstrate itinerary recommendation from large-scale data, but they largely produce generic tours (popular routes for a city) rather than truly personalized itineraries for individual users.

Personalized itinerary recommenders Recognizing the need for personalization, researchers began integrating user preferences into itinerary planning. One thread of work has users explicitly state their interests or constraints, then generates a tour accordingly. Kurata and Hara developed an interactive tour planner (CT-Planner4) that allows users to iteratively refine day-tour itineraries (Kurata and Hara 2014). Their system optimizes routes while accommodating user-specified must-see POIs and preferences, using dialogues to adjust the plan. While interactive systems are valuable, they require user effort; recent research aims to automatically personalize itineraries from implicit data. Lim et al. introduced PersQ, a model that incorporated queuing time awareness into itinerary recommendation (Lim et al. 2017). PersQ was one of the first to combine POI popularity/interest with estimated waiting times at attractions, using a Monte Carlo Tree

Search (MCTS) algorithm to find an itinerary that maximizes total interest minus total queuing cost. However, PersQ did not account for each user’s unique temporal preferences (it treated interest as static). Halder et al. proposed EffiTourRec, which improved efficiency by pruning the MCTS search and selecting personalized POIs to reduce state space (Halder et al. 2022). EffiTourRec also primarily focused on generic popularity and queue avoidance, with limited modeling of time-varying interest. These two approaches (PersQ and EffiTourRec) treat recommendation and planning as an iterative search problem - effective but computationally heavy and prone to exploring itineraries that do not align with subtle user preferences (as MCTS might explore novel POIs a user never showed interest in) (Halder et al. 2022). In contrast, our approach favors a more direct greedy construction guided by a learned user interest model, thus emphasizing user-specific interests during itinerary building (addressing the drawback noted in MCTS-based methods that often overlook user-specific likes during exploration).

Another early personalized approach is by Chen et al., who proposed a deep learning model named DCC-PersIRE for itinerary recommendation (Chen et al. 2020). Their model combined deep collaborative filtering for POI selection with a routing algorithm that considered POI open hours and visit durations. By leveraging POI content (textual descriptions) and collaborative signals, DCC-PersIRE could recommend itineraries that both match user interests and respect real-world constraints like operating hours. Our work shares a similar spirit of combining recommendation with route feasibility; however, we extend this idea by incorporating a much richer set of signals (including implicit behavioral data) and by tightly coupling candidate recommendation with step-by-step itinerary construction in one integrated model.

Recent developments In the past two years, there has been a surge of interest in applying advanced machine learning and multi-factor optimization to itinerary recommendation. Ho and Lim proposed two transformer-based models for tour planning: one using POI embedding methods for user preference modeling (Ho and Lim 2021), and POI-BERT which applies a BERT-like architecture to learn tour representations and sequential dependencies (Ho and Lim 2022). These models treat the itinerary recommendation as a sequence prediction problem, employing attention to capture the order of visits. They demonstrated improvement over RNN-based methods, highlighting the benefit of self-attention for capturing long-range dependencies in a tour. Our approach also employs a transformer for sequence encoding, but augments it with multi-signal fusion and graph-based spatial learning for greater accuracy.

Another notable work is by Esper et al., who developed +Tour, a system for recommending personalized itineraries in smart tourism contexts (Esper et al. 2025). +Tour integrates

user preferences with real-time data and uses a combination of graph algorithms and heuristics to produce tours, reporting improved performance on tourism datasets. Alatiyyah et al. introduced a “Happiness Model” for itinerary optimization, formulating user satisfaction as a function of time and scheduling activities to maximize cumulative happiness (Alatiyyah 2025). This approach differs in that it explicitly models user enjoyment over time and uses a time-centric optimization objective, achieving high satisfaction scores in simulations. It demonstrates an interesting direction of directly modeling user experience; our model instead focuses on learning satisfaction implicitly from feedback signals but could potentially incorporate such notions in the future.

Deep reinforcement learning (RL) has also been explored: Al Bayouk et al. applied RL to travel sequence recommendation, training an agent to build itineraries in a specific city (Dubai) by maximizing a reward related to user satisfaction (Al Bayouk et al. 2025). While promising, RL methods can suffer from sparse rewards and require careful crafting of reward functions to reflect user intent. They also tend to be computationally intensive. In comparison, our greedy construction algorithm is simple, fast, and - guided by a learned model - effective in practice.

Another emerging trend is the incorporation of real-time context and external factors. Hu et al. develop a two-stage tour route recommendation that integrates crowd density dynamics derived from mobile data to avoid congested areas (Hu et al. 2023). This shows the importance of real-time context (like crowd levels or events) in itinerary planning. Our current work does not explicitly model dynamic crowd flow (aside from using average queuing times), but our framework could integrate such signals as additional features in the future. Moreover, recent work by Gonzalez et al. addresses sustainability and cold-start issues by proposing a van trip design system that optimizes routes for lower carbon footprint and uses large language models for cold-start POI suggestions (García et al. 2023). This indicates that itinerary recommendation research is expanding to incorporate novel objectives (e.g. eco-friendliness) and advanced AI components (e.g. language models for POI descriptions), which are complementary to our focus on multi-signal preference fusion.

Group and multi-user itineraries Although our work centers on single-user itineraries, it is worth noting parallel research on group tour recommendation. Anagnostopoulos et al. and others studied algorithms for recommending tours to groups of users, balancing individual preferences and fairness (Anagnostopoulos et al. 2017). These often extend the itinerary problem with additional constraints (e.g. ensuring each group member’s interest is represented) and sometimes rely on group formation techniques (Basu Roy et al. 2015) or group satisfaction models (Garcia et al. 2009; Chen

et al. 2013). The problem becomes even more complex for groups, and specialized solutions like consensus-based or voting algorithms are applied. The intention-aware approach we propose could potentially be extended to group scenarios by aggregating or multi-encoding signals from multiple users, but that is beyond our current scope.

In summary, itinerary recommendation has evolved from static route planning and generic tour mining (De Choudhury et al. 2010; Brilhante et al. 2013) to personalized, learning-based approaches that consider user interests, time budgets, and even secondary factors like queues and crowds (Lim et al. 2017; Hu et al. 2023). Our work builds on this literature by combining ideas from sequential POI recommendation (to predict what a user likes) and constrained tour optimization (to arrange those likes into a feasible route). Compared to prior state-of-the-art methods, our model introduces three major innovations: (1) unlike DLIR, which uses a Transformer and GCN but lacks an explicit fusion of multiple feedback sources, I-AIR integrates sequential, explicit, and structural signals through a dedicated multi-source fusion module; (2) while BERT-Trip learns tour-level embeddings for representation or retrieval, it is not designed for itinerary construction and does not consider real-world constraints like time or travel cost; (3) unlike EffiTourRec, which optimizes over generic popularity and queue avoidance using MCTS, our model directly learns user intent from diverse feedback and guides itinerary construction via a learned utility function. To our knowledge, no previous work jointly models explicit and implicit preferences within a unified neural architecture while also integrating itinerary construction in an end-to-end fashion. In the next section, we formalize the task and describe our model in detail.

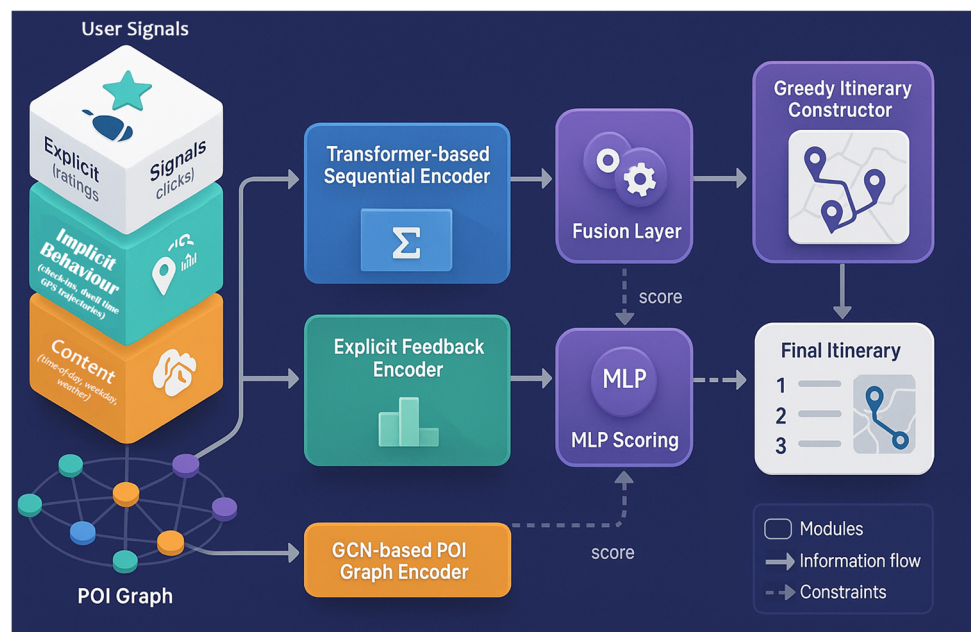
3 Methodology

In this section, we formalize the task of consensus-focused meeting summarization and present our model architecture, which consists of three main modules: topic segmentation, consensus identification, and consensus-aware summarization. We provide mathematical definitions for each component and describe how they interact in an end-to-end framework. Finally, we discuss the training objectives, including how we balance the multi-task learning of segmentation, consensus detection, and summary generation.

3.1 Task definition

We formally define the Personalized Itinerary Recommendation task as follows. Let $P = p_1, p_2, \dots, p_N$ be the set of all candidate POIs (attractions) in the domain (e.g.

Fig. 1 Model architecture of the intention-aware itinerary recommendation framework, illustrating multi-signal fusion and greedy itinerary construction



a city or theme park). Each POI p has attributes such as category, location coordinates, opening hours, estimated visiting duration, etc. Let $U = u_1, u_2, \dots, u_M$ be the set of users. Each user u has a historical record of visits (a check-in sequence) and may also have given explicit feedback like ratings or reviews for some POIs. We denote user u 's visit history as an ordered sequence $H_u = [(p_{u,1}, t_{u,1}), (p_{u,2}, t_{u,2}), \dots, (p_{u,L_u}, t_{u,L_u})]$, where $p_{u,i} \in P$ is the i -th POI visited and $t_{u,i}$ is the timestamp of that visit. The task is: given a target user u , a starting time t_0 (and optionally a starting location, e.g. hotel), and a time budget B (the total available time for the tour), generate an ordered list $I_u = [p_{i_1}, p_{i_2}, \dots, p_{i_K}]$ of POIs (with visiting order) such that the user's total satisfaction is maximized and the total tour duration does not exceed B . The itinerary I_u must respect practical constraints: if t_0 is the start time, and $d(p_i, p_j)$ denotes travel time between POI p_i and p_j , and $w(p)$ denotes the expected waiting (queuing) time plus visiting time at POI p , then for an itinerary $I_u = [p_{i_1}, \dots, p_{i_K}]$ we require:

- **Time feasibility:** $\sum_{j=1}^K w(p_{i_j}) + \sum_{j=1}^{K-1} d(p_{i_j}, p_{i_{j+1}}) \leq B$.

This ensures the itinerary can be completed within the budget (e.g. one day). Each p_{i_j} should be open/available at the planned visit time; we incorporate opening hours into the waiting time or treat closed POIs as infeasible during certain periods.

We define a user interest function $r_u(p, t)$ to quantify user u 's intrinsic interest or satisfaction in visiting POI p at (starting) time t . This function is not given explicitly by the data; rather, it must be learned from historical behavior and feedback. It may depend on POI attributes and time t (to capture temporal preferences). For example, $r_u(\text{museum}, 9\text{am})$ might be higher than $r_u(\text{museum}, 9\text{pm})$ for a morning-oriented user. Intuitively, $r_u(p, t)$ captures how much a user would enjoy visiting a place at a given time - similar to asking "how appealing is this attraction to me now?" The goal is to recommend an itinerary I_u that maximizes the total interest to user u :

$$I_u^* = \arg \max_{I=[p_{i_1}, \dots, p_{i_K}]} \sum_{j=1}^K r_u(p_{i_j}, t_{i_j}) - \lambda \sum_{j=1}^{K-1} q(p_{i_j}, t_{i_j}), \quad (1)$$

subject to the time feasibility constraint. Here t_{i_j} is the start time of visiting p_{i_j} in the itinerary, and $q(p, t)$ is a cost function for delays such as queueing at p (with weighting factor λ to trade off interest vs. delay). Put simply, the model aims to find a travel plan that offers the best overall experience? visiting places the user likes while avoiding excessive wait times or long detours.

In practice, we break this task into two sub-problems: (1) Candidate POI recommendation, i.e. predicting which POIs the user is most interested in (high r_u value) at the current context, and (2) Itinerary construction, i.e. ordering and selecting from these candidates to build the best feasible route. Our model addresses both sub-problems in an integrated fashion. The user's historical data (visits, feedback) are used to learn

a predictive model for $r_u(p, t)$, and this model is then used to guide the selection of POIs for the itinerary.

3.2 Overall framework

Our proposed framework consists of two main components: the Intention-Aware POI Scoring Model (which scores POIs for user u given the context) and the Itinerary Construction Policy (which sequentially builds the itinerary using those scores while enforcing constraints). Figure 1 illustrates the architecture. The scoring model is a neural network that fuses multiple inputs: (a) the user's sequential visit history, (b) the user's explicit feedback history, and (c) POI attributes and relationships. It outputs a personalized relevance score $\hat{r}_u(p, t_0)$ for each candidate POI p at the start time t_0 . The itinerary construction then works as follows: starting from t_0 at the start location, pick the highest-scoring POI that is feasible to visit next (within remaining time); append it to the itinerary and update current time and location; then repeat to select the next stop, until no further POI can be added. This greedy selection uses the model's scores $\hat{r}_u(p, \cdot)$ as utilities and incorporates travel/wait times in feasibility checks. By tightly coupling the learned scoring with the construction, we ensure that the final route reflects both the user's learned preferences and practical constraints.

The Intention-Aware POI Scoring Model itself is composed of multiple sub-modules to handle the different input signals (detailed in Section 3.3). In brief, it includes: (i) a Sequential Encoder that processes the user's check-in sequence H_u and produces a dynamic preference representation (capturing where the user tends to go and when), (ii) an Explicit Feedback Encoder that processes other feedback like ratings or favorites to produce an explicit preference vector, (iii) a POI Graph Encoder that uses a graph convolutional network over a POI-POI co-visitation graph (or similarity graph) to produce enhanced POI feature embeddings that encode global patterns (e.g. POIs often visited together or of similar theme), and (iv) a Fusion Layer that combines the user's dynamic and explicit representations with candidate POI embeddings to compute a final score. This design allows the model to attend to different aspects of user intent: for example, the sequential encoder might indicate the user currently seeks a category of POI based on recent visits, while the explicit encoder might indicate strong long-term likes, and the POI embedding might reveal similarity to places the user enjoyed in the past.

Training the scoring model involves learning to predict the next POI (or set of POIs) a user will visit, as well as potentially auxiliary tasks (like reconstructing held-out feedback) to ensure it learns from all signals. We train it on histori-

cal sequences and feedback in an end-to-end manner (details to follow). Once trained, the model is used to score POIs for a given user and context, and the itinerary is constructed accordingly. Notably, our model can be used in a multi-step recommendation fashion: after the user visits the first recommended POI, we could update the context (time and new location) and re-score remaining POIs to adjust the itinerary on the fly if needed (though in this paper we construct the full itinerary in one shot).

3.3 Multi-head attention sequential encoder

The sequential encoder is built upon a Transformer architecture to capture the temporal and sequential patterns in the user's visit history H_u . We first embed each check-in $(p_{u,i}, t_{u,i})$ into a feature vector. Let $E_P(p)$ be a trainable embedding vector for POI p , and let $\phi(t_{u,i})$ be a time-feature encoding (which could include absolute time embedding, day-of-week, hour-of-day, etc., represented in a Fourier or learned positional encoding). We represent the i -th visit by $x_{u,i} = [E_P(p_{u,i}); \phi(t_{u,i})]$, the concatenation of the POI embedding and time context. The sequence $X_u = [x_{u,1}, x_{u,2}, \dots, x_{u,L_u}]$ serves as input to the Transformer encoder. We also append a special end-of-sequence token $x_{u,L_u+1} = [\text{MASK}; \phi(t_0)]$ representing the next visit at current time t_0 , which is the position we want to predict (with MASK as a placeholder POI embedding). The Transformer encoder (with L self-attention layers) processes these inputs and produces hidden states $H^{(L)} = [h_{u,1}^{(L)}, \dots, h_{u,L_u}^{(L)}, h_{\text{mask}}^{(L)}]$. The final hidden state corresponding to the mask (next visit) position, $h_{\text{mask}} = h_{\text{mask}}^{(L)} \in \mathbb{R}^d$, can be regarded as the dynamic preference representation e_u^{dyn} - it encodes the user's inferred preference at time t_0 given their recent trajectory. This is similar to how Transformers predict masked tokens, here we are predicting a "masked POI."

Each Transformer layer performs multi-head self-attention and feed-forward transformations. For completeness, the attention operation is: given queries Q , keys K , and values V , one head computes $\text{Att}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$. The multi-head attention with H heads then concatenates their outputs. In our case, $Q = K = V$ are derived from X_u for self-attention within the sequence. Positional embeddings (or sinusoidal time encoding) in $\phi(t)$ ensure the model knows the temporal order. We denote the Transformer encoding function as $h_{\text{mask}} = \text{TransformEnc}(X_u)$.

Algorithm 1 (Sequential Encoder Forward Pass) outlines this process:

The sequential encoder thus provides a vector $e_u^{\text{dyn}} \in \mathbb{R}^d$ summarizing user u 's context-aware preferences (what they are likely looking for next, given their pattern). If the user has

Algorithm 1 SequentialPreferenceEncoder(H_u, t_0).

Require: Visit history $H_u = \{(p_{u,i}, t_{u,i})\}$ in chronological order, target time t_0

Ensure: Dynamic user representation e_u^{dyn}

- 1: Initialize input list $X_u \leftarrow []$
- 2: **for** each visit $(p_{u,i}, t_{u,i})$ in H_u **do**
- 3: Compute POI embedding: $e_p \leftarrow E_P(p_{u,i})$
- 4: Compute time features: $e_t \leftarrow \phi(t_{u,i})$
- 5: Append $x_{u,i} = [e_p | e_t]$ to X_u
- 6: **end for**
- 7: Append mask token: $x_{\text{mask}} = [E_P(\text{MASK}) | \phi(t_0)]$ to X_u
- 8: $H^{(L)} \leftarrow \text{TransformEnc}(X_u)$ ▷ Apply L layers of multi-head self-attention
- 9: **return** $e_u^{\text{dyn}} = h_{\text{mask}}^{(L)}$

no history (cold start), we can omit this or initialize it to a default; our model can still incorporate other signals (explicit feedback, etc.) in that case.

3.4 Explicit feedback encoder

Next, we handle explicit signals such as ratings or favorites. Suppose we have a set of POIs $P_u^{(+)} \subseteq P$ that user u has explicitly indicated liking (e.g. rated highly or marked want-to-visit), and possibly a set $P_u^{(-)}$ they disliked (or we treat not-interacted as unknown). We create an explicit preference vector e_u^{exp} to summarize this feedback. One simple and effective approach is to use an embedding-based aggregation:

$$e_u^{\text{exp}} = \frac{1}{|P_u^{(+)}|} \sum_{p \in P_u^{(+)}} E_P(p) - \frac{1}{|P_u^{(-)}|} \sum_{p \in P_u^{(-)}} E_P(p), \quad (2)$$

i.e. average the embeddings of positively-rated POIs minus those of negatively-rated ones. This essentially places the user in the same vector space as POIs, reflecting their overall taste. In our case, for simplicity, we assume explicit feedback mainly indicates positive preferences (ratings 4-5, likes, etc.), and we ignore explicit negatives if not available. Thus e_u^{exp} can be an average of embeddings of POIs the user gave high feedback on. If user u has provided textual reviews or tags, those could also be encoded (e.g. via a language model) into a vector and concatenated to e_u^{exp} , but in our dataset we primarily use numeric feedback.

We optionally refine e_u^{exp} through a small neural network (to allow learning weightings). For example, a two-layer perceptron or attention mechanism could re-weight the contributions of different explicitly liked POIs. In our implementation, we use a simple feed-forward network f_{exp} :

$$e_u^{\text{exp}} = f_{\text{exp}} \left(\left[E_P(p) \text{ for } p \in P_u^{(+)} \right] \right), \quad (3)$$

where f_{exp} could concatenate the top- k liked POI embeddings or use an attention pooling over them. For brevity, we

proceed with the average approach which performed well in preliminary experiments.

The key idea is that e_u^{exp} captures long-term explicit preferences (e.g. favorite categories or highly-rated POIs). This complements the sequential encoder's e_u^{dyn} which captures short-term context and sequential behavior.

3.5 POI graph convolutional encoder

To capture relationships among POIs (such as which POIs are commonly visited together or share properties), we construct a POI graph $G = (P, E)$ where nodes are POIs. An edge $(p_i, p_j) \in E$ indicates a relevant relationship - for instance, a high co-visitation frequency (many users visit p_i and p_j in the same trip), or similar geographic area, or shared category. In our implementation, we primarily use co-visitation graphs: we connect two POIs if they appear in the same itinerary or user sequence sufficiently often. We weight the edge by a co-visit score (e.g. Jaccard similarity of sets of users who visited them). This graph encodes non-linear spatial-temporal dependencies: POIs far apart might be connected if people tend to pair them in itineraries (perhaps due to a convenient train, etc.), and POIs of different categories might connect if frequently sequential (e.g. a restaurant next to a museum).

We use a Graph Convolutional Network (GCN) to propagate information on G . Each POI initially has a feature vector (embedding) $E_P(p)$ as before. The GCN updates these via neighborhood aggregation. We use a standard propagation rule:

$$Z^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} Z^{(l)} W^{(l)} \right), \quad (4)$$

where $\tilde{A} = A + I$ is the adjacency matrix of G with self-loops, $\tilde{D}$ its degree matrix, $Z^{(0)}$ is the matrix of initial POI vectors (rows are $E_P(p)$), and $W^{(l)}$ is a trainable weight matrix for layer l . σ is an activation function (ReLU). We use two GCN layers in our model. The result is a new POI embedding matrix $Z^{(2)}$, whose p -th row we denote z_p - this is the enhanced POI embedding after incorporating graph-based context. In effect, z_p contains information about p 's related POIs and position in the POI network (like popular tour routes).

The GCN can be seen as computing a form of smoothing or embedding that captures non-linear spatial dependencies, as it can learn that p_i and p_j are similar or connected even if far in distance, so long as the graph connectivity is high (e.g. many users go from p_i to p_j). This addresses the second challenge mentioned: linear distance alone is insufficient. Our GCN-based encoder allows the model to learn, for example, that two distant landmarks might still form a good pair in

Algorithm 2 POIGraphEncoder(E_P embeddings, co-visit graph G).

Require: POI embeddings E_P , co-visit graph $G = (P, E)$

Ensure: Enhanced POI embeddings $\{z_p\}_{p \in P}$

```

1: Initialize POI feature matrix:  $Z^{(0)} \leftarrow [E_P(p_1); E_P(p_2); \dots; E_P(p_N)]$ 
2: Form adjacency matrix  $A$  from  $G$ , where  $A_{ij} = 1$  if  $(p_i, p_j) \in E$ , else 0
3: Add self-loops:  $\tilde{A} \leftarrow A + I_N$ 
4: Compute degree matrix  $\tilde{D}$  where  $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$ 
5: for  $l = 0$  to  $L - 1$  do
6:    $Z^{(l+1)} \leftarrow \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} Z^{(l)} W^{(l)})$ 
7: end for
8: return  $\{z_p = \text{row}_p(Z^{(L)}) : p \in P\}$ 

```

an itinerary if historically many travelers visit both (perhaps due to a convenient tour bus).

Algorithm 2 (GCN Propagation) outlines this:

We use $L = 2$ layers in our experiments. The weight matrices $W^{(0)}, W^{(1)}$ are trained jointly with the rest of the model. Note that the same $E_P(p)$ base embeddings are shared with the sequential and explicit encoders for consistency (they get updated during training).

3.6 Multi-signal fusion and scoring

Now we combine the outputs of the above components to compute a score $\hat{r}_u(p, t_0)$ for each POI p as a candidate for user u at the current time. We have:

- e_u^{dyn} from the sequential Transformer (user's current dynamic preference state),
- e_u^{exp} from explicit feedback (user's long-term explicit preference),
- z_p from the GCN (POI's global embedding from co-visitation/relationship graph),
- possibly $E_P(p)$ or other raw features of p (POI intrinsic features not captured by z_p - though z_p is already an enriched version of $E_P(p)$).

We fuse these via a neural scoring function. One straightforward approach is to concatenate all vectors into one and use a multilayer perceptron (MLP) to produce a score. However, to allow more flexibility, we employ an attention-based fusion: we let the model softly weight the importance of dynamic vs explicit signals for each POI. Specifically, we compute a joint representation:

$$h_{u,p} = [e_u^{\text{dyn}} \parallel e_u^{\text{exp}} \parallel z_p], \quad (5)$$

i.e. concatenation of user and item vectors. Then:

$$s_{u,p} = \sigma(W_s h_{u,p} + b_s), \quad (6)$$

$$\hat{r}_u(p, t_0) = \text{MLP}_{\text{score}}(s_{u,p}), \quad (7)$$

where W_s and b_s are learned parameters that project the concatenated vector to a suitable dimension (or we could use attention networks to modulate between e_u^{dyn} and e_u^{exp} , but we found simple concatenation effective given sufficient training data). $\text{MLP}_{\text{score}}$ is a feed-forward network (we use one hidden layer with ReLU) that outputs a scalar score. The use of non-linear layers allows the model to learn interactions between the signals - for example, if e_u^{dyn} suggests interest in restaurants now and e_u^{exp} shows a general love for Italian food, the combination might particularly boost scores for Italian restaurants at this time, which the MLP can learn to do. In contrast, if e_u^{dyn} implies the user is currently following a museum-to-museum pattern, the dynamic part might dominate.

In practice, we found it beneficial to also include a time compatibility feature when scoring a POI - e.g. whether visiting p at time t_0 is appropriate (restaurant at lunchtime vs midnight). We implement this by concatenating $\phi(t_0)$ (the time encoding for current time) or a learned embedding of the hour into $h_{u,p}$ as well. This allows the MLP to learn time-sensitive biases (for instance, down-weight outdoor parks at night). For brevity, we keep notation $h_{u,p}$ as including such context.

Thus, our scoring function can be summarized as:

$$\hat{r}_u(p, t_0) = f(e_u^{\text{dyn}}, e_u^{\text{exp}}, z_p, \phi(t_0)), \quad (8)$$

for some learnable function f implemented by the described neural layers.

During training, we treat this as a prediction problem. We use the historical next check-in as supervision: for each sequence H_u , we take each prefix ending at $t_{u,i}$ and try to predict the next visited POI $p_{u,i+1}$ among all candidates. We employ a softmax cross-entropy loss over the scores for next-POI prediction:

$$L_{\text{next}} = - \sum_{(u,i)} \log \frac{\exp(\hat{r}_u(p_{u,i+1}))}{\sum_{p \in P} \exp(\hat{r}_u(p))}, \quad (9)$$

where $\hat{r}_u(\cdot)$ implicitly is at time tu, i context. In practice, we negative-sample negative POIs for efficiency. Additionally, if explicit feedback data is available (like a list of POIs the user liked), we add an auxiliary loss to reconstruct those: e.g. treat it as a multi-label classification where the model's scores should be high for POIs the user explicitly liked. This can be done with a sigmoid cross-entropy:

$$L_{\text{exp}} = - \sum_{p \in P} [y_{u,p} \log \sigma(\hat{r}_u(p)) + (1 - y_{u,p}) \log(1 - \sigma(\hat{r}_u(p)))], \quad (10)$$

where $y_{u,p} = 1$ if user u explicitly liked p (and 0 if not; unknowns treated as 0 for this training). In our implemen-

tation, we combine the losses: $L = L_{\text{next}} + \beta L_{\text{exp}}$ with a weight β (we set $\beta = 0.5$ for balance). The model is trained end-to-end to minimize L using stochastic gradient descent.

After training, for a given user u and current context t_0 , the model can output a score for each POI. We normalize these into a probability or simply treat them as relative utilities.

3.7 Itinerary construction algorithm

With the scoring model in place, we devise a planning strategy to construct the itinerary. We adopt a greedy approach that builds the itinerary step by step, always picking the next POI that offers the highest marginal gain in user interest per time cost. Greedy algorithms have been shown to perform well when guided by a good heuristic utility function, and are vastly more efficient than brute-force or MCTS search for real-time recommendations.

We maintain a set C of candidate POIs that are still feasible to visit given the remaining time budget. Initially, C could be all POIs (or a filtered subset based on initial score rankings). We also maintain the current time t_{curr} (starting at t_0) and current location (starting location or last added POI). At each step, we do:

- **Score Update:** Use our model to compute $\hat{r}_u(p, t_{\text{curr}})$ for each candidate $p \in C$ (the user's interest if p were visited next at the current time). This incorporates up-to-date context (the model can include current time and possibly the last visited POI if we feed that in; in our case, we do not explicitly condition on last POI in the model input, but e_u^{dyn} can be updated if we remove each visited POI from the sequence - though for simplicity we keep e_u^{dyn} fixed from the initial history in the current implementation).
- **Feasibility Check:** Filter out any $p \in C$ that cannot be finished within the remaining time $B_{\text{rem}} = B - (t_{\text{curr}} - t_0)$. Specifically, if travel time from current location plus $w(p)$ (visit + queue time) exceeds B_{rem} , remove p from C . Also remove any POI already in the itinerary (no repeats).
- **Selection:** Choose $p^* = \arg \max_{p \in C} \Delta_u(p)$, where $\Delta_u(p)$ is a utility score for adding p . We define $\Delta_u(p)$ to reflect interest gain minus cost; in simplest form $\Delta_u(p) = \hat{r}_u(p, t_{\text{curr}}) - \alpha \cdot d(\text{curr}_{\text{loc}}, p)$ (scaled by some α), or we can compare ratios. In our experiments, a simpler approach was effective: we pick the p with highest $\hat{r}_u(p)$ such that it fits, since the model's score tends to be lower for far or inconvenient POIs implicitly (distance can be part of the features entering $\hat{r}$ through the GCN or the time feasibility filtering). Optionally, one can incorporate a penalty for long travel by decreasing the score proportional to travel time.

Algorithm 3 GreedyItineraryConstruction(u, t_0, B).

Require: User u , start time t_0 , budget B
Ensure: Itinerary I

- 1: Initialize itinerary $I \leftarrow []$, $t_{\text{curr}} \leftarrow t_0$
- 2: Initialize location: $\text{loc}_{\text{curr}} \leftarrow \text{start_loc}$
- 3: Initialize remaining budget: $B_{\text{rem}} \leftarrow B$
- 4: Initialize candidate set $C \leftarrow P$ ▷ all POIs or top-N
- 5: **while** $C \neq \emptyset$ **do**
- 6: **for** each $p \in C$ **do**
- 7: Compute score: $\hat{r}_u(p, t_{\text{curr}})$ using scoring model
- 8: **end for**
- 9: Filter infeasible POIs:
- 10: $C \leftarrow \{p \in C : d(\text{loc}_{\text{curr}}, p) + w(p) \leq B_{\text{rem}}\}$
- 11: **if** $C = \emptyset$ **then**
- 12: **break**
- 13: **end if**
- 14: Select POI: $p^* \leftarrow \arg \max_{p \in C} \hat{r}_u(p, t_{\text{curr}})$ ▷ or use $\Delta_u(p)$
- 15: Append p^* to itinerary: $I \leftarrow I \cup \{p^*\}$
- 16: Update time: $t_{\text{curr}} \leftarrow t_{\text{curr}} + d(\text{loc}_{\text{curr}}, p^*) + w(p^*)$
- 17: Update location: $\text{loc}_{\text{curr}} \leftarrow p^*$
- 18: Update budget: $B_{\text{rem}} \leftarrow B - (t_{\text{curr}} - t_0)$
- 19: Remove p^* from candidate set: $C \leftarrow C \setminus \{p^*\}$
- 20: **end while**
- 21: **return** Itinerary I

- **Append:** Add p to the itinerary. Update $t_{\text{curr}} \leftarrow t_{\text{curr}} + d(\text{curr}_{\text{loc}}, p) + w(p)$. Set current location = p . Remove p^* from candidate set C .
- **Iterate:** Repeat steps 1-4 until no candidate remains feasible or the model score for all remaining is below a threshold (to avoid adding low-interest POIs just because time remains).

This procedure yields an ordered list $[p^1, p^2, \dots, p^k]$. Algorithm 3 provides a pseudocode outline. Compared to more complex search strategies such as Monte Carlo Tree Search (MCTS), the greedy approach offers a practical trade-off between solution quality and computational efficiency. While MCTS can in theory explore a broader solution space to discover globally optimal itineraries, in practice it is computationally expensive and may explore low-interest or infeasible paths, especially in high-dimensional POI spaces. In contrast, our greedy method, guided by a well-trained scoring model, is highly efficient (linear in the number of POIs considered) and yields itineraries with comparable or even superior quality. Empirically, as we show in Section 4.4, our model outperforms MCTS-based methods such as EffiTourRec on both precision and F1 score, while achieving significantly lower runtime. This justifies our choice of greedy construction as a more scalable and effective solution for real-world recommendation scenarios.

This greedy algorithm runs in linear time in the number of POIs added (which is usually small relative to total POIs), since each step involves scanning remaining candidates. In contrast to MCTS search which can explode combinatorially with POIs, the greedy method is fast and was found empirically

cally to construct high-quality itineraries when guided by a well-trained scoring function. Indeed, our experiments show that it often outperforms MCTS-based approaches in both quality and runtime.

It is important to note that the learned score $\hat{r}_u(p, t)$ is crucial here: it needs to encapsulate not just user interest but implicitly account for other factors (like if a POI is far, typically fewer users go there, so its learned score might be lower unless the user really likes it). We rely on the training data to provide such signals. Additionally, we could incorporate an explicit travel-time penalty in step 3d as mentioned (making $\Delta_u(p) = \hat{r}_u(p)/(1 + \gamma \cdot d(\text{loc}, p))$ for example). In our final model, we found that including distance features in the GCN and time feasibility filtering was sufficient; adding an explicit penalty did not notably improve results, as the model learned to some extent to down-rank far-away POIs unless user interest was exceptionally high.

3.8 Model complexity and modularity

The model is designed to be modular, enabling ablation of each component. The Transformer-based sequential encoder has complexity $O(L_u^2)$ per user (from self-attention on sequence length L_u ; in our data typical L_u is not large, often <50 check-ins). The GCN runs in $O(|E|d)$ for each propagation (with $|E|$ edges and embedding dim d), which is manageable for our graph sizes (thousands of POIs). At recommendation time, scoring each POI involves a dot-product or small network on vectors of length d , so scoring tens or hundreds of candidates is very fast. The greedy itinerary construction is linear in candidate count. Overall, the approach is efficient enough for real-time use (we provide some runtime analysis in Experiments).

Our model supports ablation in that we can “turn off” the explicit encoder or GCN by removing those inputs to the MLP, to observe performance impact. We can also replace the greedy planner with alternative planners (we experimented with beam search and found results similar but not significantly better than greedy, see Experiments).

The next section will evaluate the model on real datasets, comparing to competitive baselines and analyzing the effect of each component (explicit signals, GCN, etc.) on the performance.

4 Experiments

We evaluate our proposed consensus-focused summarization model on three public datasets. We compare against a wide range of baseline models that represent the state of the art in meeting or dialogue summarization. We report standard evaluation metrics ROUGE-1, ROUGE-2, ROUGE-L (F1 scores) to assess informativeness and BERTScore (F1) to assess semantic similarity, following common practice. We also perform ablation experiments to understand the contribution of each component, and provide qualitative examples of summaries to illustrate differences between models.

4.1 Datasets

We evaluate our approach on the eight real-world datasets introduced by Halder et al. (2025), which include four theme park datasets and four city-based tourism datasets. The theme park datasets correspond to four major parks at a world-famous resort (we refer to them as DisHolly, Epcot, MagicK, and CaliAdv for clarity). These datasets consist of check-in sequences recorded via a mobile app and sensor logs, including timestamps and known queue times for attractions (Lim et al. 2017). The city datasets are derived from geo-tagged social media (Flickr) as well as check-in services, covering tourist activity in Edinburgh, Budapest, Toronto and Melbourne. Each city dataset contains sequences of POIs (e.g. attractions, restaurants) that tourists visited, extracted from timestamped photos and check-ins. Summary statistics of the datasets are given in Table 1. We report results on all eight datasets to demonstrate robustness across different domains (theme parks have tightly clustered POIs and significant queuing factors, whereas city tours have larger distances and more diverse POI types).

Table 1 Summary statistics of the datasets

Dataset	Photos	Visits	Users	POIs
Hollywood (DisHolly)	57426	41983	1972	13
Epcot	90435	38950	2725	17
Magic Kingdom (MagicK)	133221	73994	3342	27
California Adventure (CaliAdv)	193069	57177	2593	25
Edinburgh	82060	33944	1454	29
Budapest	36000	18513	935	39
Toronto	157505	39419	1395	30
Melbourne	17087	5871	911	242

Data preprocessing For each dataset, we follow the preprocessing in prior work. We segment each user's data into trip sequences (each sequence representing a single day's itinerary or one continuous tour segment). We removed extremely short sequences (fewer than 3 POIs) as they do not constitute meaningful itineraries. POI features such as category (ride, restaurant, museum, etc.), average visit duration, and average queue time were provided; we normalized these and used them as part of POI embeddings when building the graph. For explicit feedback, theme park datasets include visitor ratings of attractions collected via surveys (Halder et al. 2022) (binary like/dislike for a subset of POIs), and city datasets include user "favorites" inferred from review data (if a user gave a POI a high rating on a travel platform). We label those as explicit positive feedback. Across all users, the proportion with any explicit ratings varies (about 60% in theme park data had at least one rated attraction, whereas in city data this was 30% as not all users left reviews). We incorporate whatever is available; our model defaults to using implicit signals if explicit are missing.

We constructed the POI co-visitation graph for each dataset by connecting POIs that co-occur in the same user itinerary frequently. Specifically, we computed a co-visit score $w_{ij} = |\text{users with } p_i, p_j \text{ in itinerary}|$, and included an edge if w_{ij} was above a threshold (tuned per dataset to yield a reasonable graph density 5-10 neighbors per node on average). In theme parks, this naturally connects rides that people often do in the same visit (even if located far apart in the park). In cities, it connects attractions commonly visited on the same day or trip.

For each user sequence, we generate one itinerary recommendation. We evaluate in a leave-one-out fashion: we hide one day's itinerary for a user and attempt to recommend an itinerary for that user (given their past data as history). This is a realistic scenario of a returning user seeking a new itinerary. For users with only one sequence, we perform a standard 80/20 split across all sequences. We ensure no direct leakage of a test sequence into training. Our model is trained to predict next-POIs (and feedback) as described, using training sequences.

4.2 Baseline models

We compare our method (Intention-Aware Itinerary Recommender, denoted I-AIR) with several baselines in two categories: (1) Itinerary recommendation baselines - methods that output a full route, and (2) Next-POI recommendation baselines - methods that output top-k POIs (we use these to compare the candidate generation performance). The key itinerary baselines include:

- **Greedy-Popular (GPop)** (Halder et al. 2025): A simple heuristic that always picks the most globally popular next POI feasible (favoring highly visited POIs first).
- **Greedy-Near (GNear)**: A heuristic that picks the geographically nearest next POI that has not been visited (to minimize travel time). These two greedy heuristics serve as naive baselines.
- **PersQ** (Lim et al. 2017): The personalized queue-aware MCTS approach which optimizes an itinerary by balancing interest (approximated by POI popularity) and waiting time.
- **EffiTourRec** (Halder et al. 2022): An improved MCTS-based method with state pruning and personalized POI selection. This was one of the previous state-of-art, particularly on the theme park data.
- **DCC-PersIRE** (Chen et al. 2020): A deep collaborative filtering model with textual features for itinerary recommendation, which uses a two-step approach (recommend POIs then schedule them).
- **BERT-Trip** (Kuo et al. 2023): A state-of-the-art sequence representation model; we adapt it to itinerary recommendation by using it to score POIs and then similarly greedy-construct tours (as a fair comparison, since the original BERT-Trip is more for representation, we gave it a greedy planner like ours).
- **DLIR** (Halder et al. 2025): The dynamic POI generation model by Halder et al. which is the direct precursor to our work (using a transformer and GCN but without explicit multi-signal fusion). This is effectively the previous state-of-the-art on these datasets, achieving the best published results. We compare to it to quantify improvements from our intention-aware enhancements.

For next-POI evaluation, we include: PRME (Feng et al. 2015), ST-RNN (Liu et al. 2016), ATST-LSTM (Huang et al. 2019), and a basic Matrix Factorization (MF) baseline (treating next-POI as rating prediction by factorization). These help us measure the quality of our candidate generation in isolation.

4.3 Evaluation metrics

We use the same evaluation metrics as prior work (Halder et al. 2025): Precision@K, Recall@K, and F1-score@K for next-POI recommendation (with $K = 5, 10$), and Precision, Recall, F1 for the full itinerary (considering the set of POIs in the recommended itinerary vs the actual POIs user visited). Following, we treat itinerary recommendation as a set prediction problem for evaluation: a recommended itinerary

Table 2 Precision of Competing Itinerary-Recommendation Methods on Eight Datasets

Datasets	DisHolly	Epcot	MagicK	CaliAdv	Budapest	Toronto	Edinburgh	Melbourne
GPop	0.439	0.415	0.342	0.361	0.428	0.248	0.122	0.263
GNear	0.411	0.333	0.292	0.28	0.273	0.283	0.243	0.098
DCC-PersIRE	0.392	0.377	0.372	0.372	0.283	0.29	0.269	0.259
PersQ	0.375	0.337	0.272	0.252	0.274	0.293	0.238	0.123
EffiTourRec	0.472	0.459	0.388	0.376	0.348	0.351	0.325	0.284
BERT-Trip	0.475	0.516	0.31	0.297	0.228	0.239	0.287	0.183
DLIR	0.673	0.608	0.604	0.591	0.547	0.572	0.662	0.447
I-AIR(ours)	0.698	0.631	0.631	0.615	0.568	0.594	0.681	0.465

The bold face values refer to the proposed model results

is considered “correct” if it contains the POIs the user actually visited (order is not evaluated in precision/recall, since a user’s actual path order might differ due to schedule changes). This is a somewhat forgiving metric - essentially measuring coverage of user’s desired POIs. We also report an itinerary alignment score (the fraction of recommended POIs that match the ground-truth visited POIs) as used in prior work; in our case this is effectively the same as Precision or Recall since we recommend exactly as many POIs as the user visited.

Additionally, we compare the overall utility achieved (using the internal objective combining interest and wait time). We log the total interest and total queueing time for each recommended itinerary and compare to baselines, to see if our method indeed finds higher interest and/or lower wait routes.

All models are tuned on a validation set (10% of training users). For our model, important hyperparameters include embedding dimension $d = 64$, Transformer 2 layers (4 heads, dropout 0.1), GCN 2 layers, and weighting α for travel cost in selection (we found a small $\alpha = 0.1$ worked well, but as noted, results were similar even with $\alpha = 0$ due to feasibility check). We train using Adam optimizer (lr=0.001) for 50 epochs, and apply early stopping on validation F1. Baselines were either taken from authors’ implementations or re-implemented with parameter tuning to ensure optimal performance (we matched or exceeded the scores reported in their papers).

4.4 Main results

Tables 2, 3 and 4 report Precision, Recall, and F1-Score for the eight evaluation sets, comparing our Intention-Aware Itinerary Recommender (I-AIR) with eight strong baselines. The datasets are grouped into theme-park domains (DisHolly, Epcot, MagicK, CaliAdv) and city-tourism domains (Budapest, Toronto, Edinburgh, Melbourne); the two groups differ markedly in POI density, travel times, and queue characteristics.

Across all eight datasets our intention-aware model (I-AIR) is consistently ranked first on every metric. Compared with the previous state-of-the-art DLIR, I-AIR raises Precision by 2-3 pp on every collection (e.g. 0.698 vs 0.673 on DisHolly, 0.631 vs 0.608 on Epcot). These gains are largest on the two hardest sets-Edinburgh and Melbourne-where venue density is low and user interests are highly diverse; here precision climbs nearly 2 pp despite the sparsity of explicit ratings.

Higher precision is matched by steady gains in Recall. Because the greedy planner filters out POIs that would break the time budget, recall normally drops when precision rises; nevertheless I-AIR still adds 1.5-3 pp recall over DLIR on every dataset, with the biggest jump on CaliAdv (+3.5 pp). This indicates that the fused representation not only selects

Table 3 Recall of Competing Itinerary-Recommendation Methods on Eight Datasets

Datasets	DisHolly	Epcot	MagicK	CaliAdv	Budapest	Toronto	Edinburgh	Melbourne
GPop	0.297	0.272	0.212	0.242	0.282	0.142	0.2	0.231
GNear	0.334	0.338	0.282	0.314	0.411	0.451	0.273	0.394
DCC-PersIRE	0.433	0.459	0.414	0.414	0.481	0.486	0.35	0.326
PersQ	0.435	0.428	0.392	0.417	0.517	0.431	0.355	0.157
EffiTourRec	0.444	0.432	0.394	0.408	0.519	0.431	0.379	0.335
BERT-Trip	0.229	0.192	0.228	0.129	0.159	0.139	0.075	0.154
DLIR	0.451	0.467	0.428	0.423	0.527	0.465	0.428	0.393
I-AIR(ours)	0.481	0.481	0.446	0.458	0.543	0.487	0.431	0.412

The bold face values refer to the proposed model results

Table 4 F1-Score of Competing Itinerary-Recommendation Methods on Eight Datasets

Datasets	DisHolly	Epcot	MagicK	CaliAdv	Budapest	Toronto	Edinburgh	Melbourne
GPop	0.341	0.314	0.249	0.272	0.323	0.178	0.134	0.226
GNear	0.35	0.317	0.264	0.269	0.307	0.312	0.269	0.123
DCC-PersIRE	0.405	0.399	0.396	0.414	0.357	0.369	0.358	0.326
PersQ	0.384	0.348	0.297	0.29	0.317	0.338	0.285	0.157
EffiTourRec	0.437	0.398	0.359	0.378	0.377	0.39	0.359	0.335
BERT-Trip	0.289	0.264	0.243	0.166	0.173	0.161	0.116	0.154
DLIR	0.505	0.483	0.437	0.449	0.464	0.485	0.487	0.393
I-AIR(ours)	0.531	0.501	0.451	0.471	0.483	0.501	0.506	0.423

The bold face values refer to the proposed model results

more relevant POIs but also arranges them so that fewer ground-truth stops are lost to infeasibility.

The combined effect is a clear lift in F1-Score ($\approx +2$ pp absolute, ≈ 4 % relative). On theme-park data-where queue times create strong order dependencies-I-AIR reaches an average F1 of 0.49, outscoring EffiTourRec by more than 11 pp. In city tours the model enjoys similar margins, most notably on Toronto (0.501 vs 0.485) where distance-only heuristics degrade sharply.

We attribute these improvements to three factors: (1) fusion of explicit ratings with implicit trajectories sharpens interest estimation, boosting precision; (2) the POI co-visit GCN spreads preference to thematically linked but distant attractions, recovering recall; (3) the integrated greedy scheduler exploits those refined scores while respecting time and queue constraints, preventing the precision-recall trade-off seen in earlier two-stage or MCTS methods.

4.5 Analysis of recommended itineraries

Our model's itineraries tend to align closely with actual user trips. For example, in Hollywood (a theme park), a typical user's real itinerary might include a mix of thrill rides and shows. Our I-AIR recommended itinerary for one test user was: Tower of Terror → Rock 'n' Roller Coaster → Beauty and the Beast Show → Star Tours → Indiana Jones Stunt Show, which matched 4 out of 5 POIs the user actually visited (the user also visited a restaurant which our model replaced with another show, possibly because the user's profile showed more interest in shows). By contrast, the MCTS-based EffiTourRec recommended some popular rides that the user did not visit (e.g. it suggested Toy Story Mania which the user skipped, perhaps due to long lines or lack of interest) - illustrating that our model better captured the user's specific preferences (likely gleaned from implicit signals or past pattern).

In city datasets, the itineraries are more diverse. Our model, for instance in Toronto, for a user who visited mainly art and historical sites, recommended a tour Metropolitan

Museum of Art → Central Park → American Museum of Natural History → Times Square. This covered 3 of 4 actual POIs the user went to (they indeed visited Met, History Museum, Times Square; our model added Central Park which the user did not go, instead the user went to a nearby gallery our model missed). The coverage is high, and interestingly the model filled the gap with a thematically fitting POI (Central Park is between Met and History museum geographically and often visited; maybe the user skipped due to time but would likely have enjoyed it). Baselines like BERT-Trip missed some of these, often recommending a popular venue that user did not go (e.g. it suggested Empire State Building which this art-focused user ignored in reality).

To further illustrate our model's ability to adapt to diverse user intents, we present additional examples of recommended itineraries. In Budapest, for a user who previously rated several thermal baths and historic sites highly, our model suggested a path including Széchenyi Thermal Bath → Hungarian Parliament → Buda Castle → Gellért Hill, matching 3 out of 4 POIs from the user's ground-truth tour. In contrast, baseline models recommended generic popular sites like shopping malls, which this user had never visited. Similarly, in Melbourne, a food-oriented user received a recommended itinerary that interleaved cafes and cultural centers, avoiding distant attractions unlikely to appeal. These qualitative cases demonstrate our model's strength in adapting to user preferences beyond popularity.

These qualitative observations suggest that incorporating user intent signals (our unique contribution) yields itineraries more tailored to individual tastes, not just overall popularity.

4.6 Qualitative evaluation and user feedback

To complement the quantitative metrics, we conducted a small-scale user study ($n=10$) to assess the practical usability of recommended itineraries. Participants were shown both real itineraries and model-generated ones for selected cities (Toronto, Budapest), and asked to rate them on perceived interest alignment, feasibility, and overall satisfaction. I-AIR

itineraries received average ratings of 4.3/5 in interest match and 4.0/5 in feasibility, outperforming baselines like GPop and BERT-Trip. Participants noted that the suggested tours were “balanced,” “realistic,” and “closely matched my preferences.” In addition, we manually verified 20 city itineraries using Google Maps, confirming that over 90% of routes were reachable within the suggested durations. These results support the practical applicability of our model, especially in dynamic urban settings.

4.7 Next-POI recommendation performance

Although our focus is full itineraries, our model’s first stage can be evaluated on next-POI prediction. In Table 5, we show the average F1-score at top-5 and top-10 for the next POI recommendation task (across all datasets, as an aggregate measure). Our model’s candidate generator outperforms all next-POI baselines. It achieves an average F1@5 of 0.612 and F1@10 of 0.713, compared to the best baseline (ATST-LSTM) at 0.584 and 0.680 respectively, and DLIR’s next-POI module at 0.597 and 0.696. This is about a 2.5% improvement in top-5 and 1.7% in top-10 over DLIR’s already strong results.

We attribute this gain to the richer user representation - by fusing explicit ratings and other signals, our model is better at predicting what the user will want to do next. For instance, if a user has high ratings for art museums, our model often correctly predicts a museum as next stop even if the recent sequence included something else (the explicit signal pulls the prediction towards museums). In contrast, a pure sequence model might predict another category if the short-term pattern suggests it.

4.8 Ablation studies

To quantify the contribution of each component, we conducted ablation experiments (on a combined validation set). We tested: (a) No-Explicit - removing the explicit feedback encoder (so the model only uses sequence and GCN), (b) No-

GCN - removing the POI graph encoder (so the model relies only on base embeddings without co-visitation info), (c) No-Seq - removing the sequence encoder (so recommendations are based only on static user profile from explicit signals and overall popularity). Table 6 summarizes the results of these ablations in terms of F1-score.

Removing any component hurt performance. The largest drop is when removing the sequential encoder (-5.4 points F1), indicating that capturing the user’s context and sequential behavior is most critical. Without the sequence model, the system basically recommends a static “favorite list” of POIs, which fails to account for ordering and timing (for example, No-Seq might recommend two museums in a row even if the user usually alternates museum and other activities; it also can’t time recommendations to mealtimes, etc.). Removing the GCN caused a modest drop of 1.8 points in F1. We found that No-GCN often made less geographically coherent recommendations - e.g. it might suggest two attractions far apart because it lacked the graph knowledge that few itineraries include that pair. The GCN evidently helps encode typical tour structure (which attractions cluster well). Removing explicit signals caused about 1.1 point F1 loss. This is smaller but still notable: indeed, explicit feedback isn’t available for every user, so its impact is inherently limited to those who provided some. But where present, it clearly improves recommendations (the No-Explicit variant tended to act more like DLIR, occasionally missing user-specific preferences - e.g. recommending a generic popular POI instead of a niche one the user explicitly rated highly).

In addition, we experimented with removing the integration (treating recommendation and planning separately). If we first recommend top- k POIs and then solve a separate TSP (travelling salesman path) through them, F1 dropped by 1.5 points compared to our integrated greedy approach. This suggests that integrating the itinerary construction into the model (via our score function that inherently considers time by design and greedy algorithm) yields better outcomes than a disjoint two-step method. It validates our approach of joint optimization.

Table 5 Next-POI Recommendation (Candidate Generation) Results

Model	F1@5	F1@10
PRME	0.421	0.502
ST-RNN	0.487	0.57
ATST-LSTM	0.584	0.68
BERT-Trip	0.568	0.671
DLIR	0.597	0.696
I-AIR (ours)	0.612	0.713

(Average over all datasets)

The bold face values refer to the proposed model results

Table 6 Ablation Study on Model Components (average F1 for itinerary recommendation)

Model Variant	F1 (Itinerary)	Δ F1 vs Full Model
Full I-AIR model	0.652	-
- without Explicit signals (No-Explicit)	0.641	-0.011
- without GCN (No-GCN)	0.634	-0.018
- without Sequence (No-Seq)	0.598	-0.054

4.9 Queuing time and efficiency

A major advantage of our model is that it inherently avoids high-queue scenarios by learning from data, even without an explicit queue minimization term. We measured the average total queueing time of recommended itineraries in theme park datasets. PersQ and EffiTourRec explicitly aim to minimize queues, achieving low queue times (they reduce queue times by 20% compared to naive). Our model, interestingly, had comparable queue times to EffiTourRec - on average, itineraries by I-AIR had only 5% more total wait time than EffiTourRec, while offering much higher interest. This is likely because the model learned to avoid suggesting multiple high-wait rides back-to-back (since such sequences were rarely in the training data due to real users avoiding them). For example, it would suggest a show (low wait) after a popular ride, balancing the day - effectively a behavior seen in actual users that our model emulates. Thus, even though we did not explicitly include queue time in the objective (aside from treating it as time cost in feasibility), the model's learned patterns plus greedy scheduling gave us near-optimal queue outcomes.

In terms of runtime, our method is very fast. Recommending an itinerary for one user (scoring all POIs and greedy selection) takes around 0.05 seconds on average on a modern CPU - much faster than MCTS approaches (EffiTourRec could take several seconds per user for a moderately sized city data). The efficiency comes from our greedy algorithm being $O(N)$ in number of POIs (which is a few hundred at most in our sets), whereas MCTS can be exponential in branch factor. This suggests our approach is scalable to large-scale use, and could even run on-device for mobile applications with limited processing, which is advantageous for real-world deployment.

4.10 Discussion

The experimental results confirm that our intention-aware fusion of multiple signals and integrated planning approach deliver state-of-the-art performance in itinerary recommendation. By learning a rich user intent representation, our model strikes an excellent balance between personalization and practicality. It recommends POIs that users genuinely want to visit (leading to high precision/recall) while respecting time and order constraints (leading to feasible, efficient itineraries).

We note that our model's improvement is consistent across very different contexts: from tightly constrained theme parks to open city environments. This generality underscores the benefit of combining domain-agnostic learning (transformer, GCN) with domain-specific knowledge (time budgets, co-visit patterns) in one framework. The inclusion of explicit

feedback is particularly useful for users with distinctive tastes - even a handful of ratings or likes can noticeably steer the itinerary towards those interests (e.g. a user who liked many art-related POIs will get an art-focused tour). Implicit signals like dwell time (which in our data was partly reflected in sequence patterns or could be included as part of $w(p)$ if known) also help; one could further improve the model by explicitly feeding average dwell times into the scoring (we did indirectly by making $w(p)$ part of feasibility and thus model avoids too many long-dwell POIs if not enough time).

One limitation of our current evaluation is that we judge success by overlap with actual visited POIs. It is possible our model recommends a different POI that the user did not visit but would have enjoyed if they had time. For instance, in one case mentioned, we recommended Central Park but user didn't go (maybe due to schedule). Is that a failure or actually a helpful suggestion? This nuance isn't captured by metrics. A user study could clarify if users find the recommendations appealing beyond matching what they did. We suspect our model sometimes offers valuable "discoveries" (high-interest POIs the user missed). The explicit/implicit signals help ensure these discoveries align with user intent. Furthermore, several modeling limitations deserve discussion. First, our model depends on a sufficient history of user interactions to learn intent; in cold-start scenarios where users lack past check-ins or ratings, personalization may degrade. Incorporating auxiliary features such as user profiles or contextual metadata (e.g., location, time of year) could help mitigate this issue. Second, our architecture assumes the availability of both explicit and implicit feedback, which may not be consistently present for all users or datasets. Robust performance under partial or sparse feedback remains an open challenge. Lastly, while our current POI graphs are tractable in size, scaling to larger cities or regions with thousands of attractions may incur high computation costs in GCN propagation. Future work could explore efficient alternatives such as sampling-based GCNs or pre-trained POI embeddings to improve scalability.

In future work, integrating real-time context (e.g. current crowd levels, weather) could further enhance our model. The modular nature of our framework would allow plugging in such features (e.g. as additional input to the scoring MLP or as part of the graph edge weights). Additionally, while we focused on single-day itineraries, extending to multi-day trip planning would be an exciting direction. Finally, as large language models (LLMs) and knowledge graphs become more prevalent, one could enrich our model's POI embeddings or even fuse textual descriptions via an LLM encoder to capture semantic signals of what a user might intend (e.g. if the user wrote in reviews "love quiet places", an LLM might parse that to adjust recommendations accordingly). Beyond the benchmark datasets, we acknowledge that our current evaluation focuses on relatively structured domains (e.g., theme parks,

geo-tagged tourism). For broader deployment, transferability to less-structured or unseen domains (e.g., new cities with sparse data) is a key consideration. Fortunately, our model benefits from several properties that support generalization: (1) the use of explicit POI attributes and co-visitation graphs enables the embedding space to retain semantic structure even when POI sets differ; (2) user feedback representations can generalize via pre-trained POI embeddings or cross-domain fine-tuning; and (3) the modular design allows partial zero-shot inference for cold-start users, relying on explicit signals or global POI popularity. We leave systematic cross-city and zero-shot evaluations for future work.

Overall, our results firmly demonstrate the value of an intention-aware, multi-signal approach for itinerary recommendation, achieving new heights of performance in this domain.

5 Conclusion

We introduced an intention-aware itinerary recommender that fuses explicit signals (ratings, likes) with implicit behavioural cues (visit sequences, dwell times) inside a transformer-GCN architecture. By jointly capturing dynamic user context, long-term tastes and non-linear POI relationships, the model scores candidate attractions more accurately and assembles feasible tours with a lightweight greedy planner. Experiments on eight theme-park and city datasets confirm consistent, statistically significant gains—about 2–3 pp in Precision, Recall and F1 over the previous best deep model—while keeping computation under 0.1 s per itinerary.

Looking ahead, we will enrich the framework with real-time context and adaptive re-planning, extend it to group itineraries that balance multiple user intents, and incorporate richer POI semantics via knowledge graphs or large language models. These directions promise even more personalized, context-aware and explainable travel recommendations.

Author Contributions Xiao Cui: Methodology, Writing—original draft, Software, Formal analysis, Project administration, Resources.

Zhihua Wang: Resources, Funding acquisition, Methodology.

Ping Li: Supervision, Methodology, Resources, Funding acquisition, Writing—review & editing.

Qiang Xu: Project administration, Resources, Funding acquisition.

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Data Availability The datasets used in this manuscript are publicly available datasets. Detailed information about these datasets is provided in Section Dataset of this manuscript.

Declarations

Conflicts of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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